BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 049 · Pseudocode and Trace Tables

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a diamond represent in a flowchart?

Q2. Why is primary storage needed?
A. To hold data in use
B. To display images
C. To connect network devices
D. To print documents

Q3. “Both RAM and ROM lose their contents when power is removed.” Correct this. Explain why.

Q4. A computer needs startup instructions when switched on. “ROM normally holds all unsaved user work while applications run.” Correct this. What could happen?

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I Do · Read new knowledge

Key vocabulary

Pseudocode — Readable structured instructions describing an algorithm.

Trace table — Table following variable values and outputs during execution.

Variable — Named storage whose value can change.

Assignment — Storing a value in a variable.

Dry run — Manually executing an algorithm step by step.

Logic error — A program executes but gives an incorrect result.

Condition — True/false test selecting an execution branch.

New learning

Pseudocode represents an algorithm: Pseudocode expresses steps in clear, structured language. OCR Exam Reference Language gives a consistent notation for variables, selection and output.

Record each assignment: A trace table follows execution instruction by instruction. After assignment, record the new value of the variable; do not use its old value again.

Trace conditions accurately: Evaluate an IF condition against the latest value. Only execute the true or false branch selected. Record printed output when a print statement runs.

Find the first incorrect step: Compare the table with expected values. When the first mismatch appears, correct that instruction and repeat the dry run with the same inputs.

Worked example

Trace: total = 4; total = total + 3; print(total). State total after each assignment and the final output. [3 marks]

1. The first line sets total to 4. Use the current value for later statements.

2. After total = 4, record 4 [1].

3. Next, total = total + 3 uses 4 + 3, so total becomes 7 [1].

4. print(total) outputs 7 [1], but does not change the variable.

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A decision or condition with alternative paths.

Q2. A — To hold data in use

Q3. RAM loses its contents; ROM keeps its contents because it is non-volatile.

Q4. ROM keeps firmware or startup instructions without power. These let the computer start when switched on.

I Do / We Do pages · Guided practice

Q1. First score is 5 [1]. Then score becomes 7 [1]. Output is 7 [1].

Q2. First count is 9 [1]; after subtraction it is 5 [1]. 5 > 6 is false [1]; Low is printed [1].

Q3. Records each variable's value after executed steps [1], allowing comparison with expected values [1]. The first incorrect value identifies where to inspect [1]; correct and rerun the trace [1].

You Do pages · Code First

```
storedMB = 10
storedMB = storedMB + 4
print(storedMB)
```

Expected result: 14

You Do pages · Independent questions

Q1. Records changing variable values as instructions execute [1], helping predict output/check logic [1].

Q2. Cost 8 [1], then 16 [1], then 19 [1]; output 19 [1].

Q3. Balance starts 20 [1], decreases to 14 [1]. 14 >= 15 is false [1] because 14 is below 15 [1]. The else branch executes [1], printing Low [1].

Exit Ticket · Model answer

True

Trace: score = 5; score = score + 2; print(score). Give both values and the output. [3 marks]

Worked method: First score is 5 [1]. Then score becomes 7 [1]. Output is 7 [1].

Guided 2

Trace: count = 9; count = count - 4; if count > 6 then print("High") else print("Low") endif. State count and output. [4 marks]

Worked method: First count is 9 [1]; after subtraction it is 5 [1]. 5 > 6 is false [1]; Low is printed [1].

Guided 3

A pupil expects 12 but a trace shows 10. Explain how a trace table helps find and fix the error. [4 marks]

Worked method: Records each variable's value after executed steps [1], allowing comparison with expected values [1]. The first incorrect value identifies where to inspect [1]; correct and rerun the trace [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the number printed after two instructions.

```
storedMB = 10
storedMB = storedMB + 4
print(storedMB)
```

predict

Question 1

State one use of a trace table. [2 marks] Source: Original J277-style teaching question, not an official OCR past paper.

Question 2

Trace: cost = 8; cost = cost * 2; cost = cost + 3; print(cost). State cost after each assignment and the output. [4 marks] Source: Original J277-style teaching question, not an official OCR past paper.

Question 3

Trace: balance = 20; balance = balance - 6; if balance >= 15 then print("Enough") else print("Low") endif. Explain the branch and output, including the changes to balance. [6 marks] Source: Original J277-style teaching question, not an official OCR past paper.

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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

The latest value of a variable is used when tracing the next instruction.